

Project ATLAS

Threat Hunt Report — TH-2026-001

Post-Incident Proactive Hunt: Cloud Identity Abuse & Endpoint Telemetry Gap

Hunt ID:	TH-2026-001	Analyst:	Ivan Yamoah Baoakye
Hunt Period:	4–5 July 2026	Trigger:	Post-incident — Incident ID 2 closure
Environment:	VANPASSSC200 law-soc-atlas DESKTOP-STAR-0179	Status:	Closed

1. Executive Summary

This threat hunt was initiated after closure of Incident ID 2 to determine: (a) whether attacker activity beyond what Entra ID Protection detected had persisted undetected on the endpoint, and (b) whether the Log Analytics workspace contained sufficient telemetry to support retrospective endpoint hunting. Key finding: the AMA telemetry pipeline was confirmed completely non-functional — both SecurityEvent and Sysmon Event tables contained 0 records. Four of five simulated attack techniques were entirely invisible. The cloud identity hunt surfaced no additional attacker activity beyond the 5 sign-in events already flagged by Entra ID Protection.

2. Hunt Hypotheses & Queries

H1: Brute force pre-cursor activity (T1110.001)

Table: SecurityEvent

Query (representative):

```
SecurityEvent
| where EventID == 4625
| where TimeGenerated >= ago(7d)
| summarize FailedAttempts=count(), FirstSeen=min(TimeGenerated),
  SourceIPs=make_set(IpAddress) by IpAddress, Computer
| where FailedAttempts > 10
| order by FailedAttempts desc
```

Expected	Multiple 4625 events from 10.0.2.15 (Kali) — brute force spike pattern
Actual	0 records — SecurityEvent table empty (AMA pipeline failure)
Status	■ Cannot hunt — no endpoint data

H2: Encoded PowerShell execution (T1059.001 + T1027)

Table: Event (Sysmon)

Query (representative):

```
Event
| where Source == "Microsoft-Windows-Sysmon"
| where EventID == 1
| where TimeGenerated >= ago(7d)
// Parse XML, filter CommandLine has "-enc"
| where tostring(EventData) has "-enc"
| project TimeGenerated, Computer, EventData
```

Expected	Process create event with powershell.exe -enc in command line
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Actual	0 records — Sysmon Event table empty (AMA pipeline failure)
Status	■ Cannot hunt — no endpoint data

H3: Tor / anonymous proxy sign-ins beyond known scope (T1090.003)

Table: SigninLogs

Query (representative):

```
SigninLogs
| where TimeGenerated >= ago(7d)
| where RiskEventTypes has "anonymizedIPAddress"
| summarize SignInCount=count(), SuccessCount=countif(ResultType=="0"),
    Locations=make_set(Location)
    by UserPrincipalName, IPAddress
| order by SignInCount desc
```

Expected	5 known events; hunt looking for additional lateral scope
Actual	2 users (admin + soc-tor-test), 5 events, 2 Tor IPs — matches incident scope exactly. No lateral spread.
Status	■ Hunted — no additional scope found

H4: Persistence mechanisms post-compromise (T1053 / T1136)

Table: SecurityEvent + Sysmon

Query (representative):

```
SecurityEvent | where EventID in (4720, 4698) | where TimeGenerated >= ago(7d)
// + Sysmon EventID 12/13 registry Run key checks
```

Expected	New accounts (4720), scheduled tasks (4698), registry run keys (Sysmon 12/13)
Actual	0 records — cannot confirm or deny persistence (telemetry gap)
Status	■ Cannot hunt — no endpoint data

3. Hunt Results Summary

Hypothesis	Technique	Table	Records	Finding
H1: Brute force	T1110.001	SecurityEvent (4625)	0	Telemetry gap — AMA pipeline failed
H2: Encoded PS	T1059.001+T1027	Event/Sysmon (EID 1)	0	Telemetry gap — Sysmon not forwarding
H2b: Proc create	T1059.001	SecurityEvent (4688)	0	Telemetry gap — AMA pipeline failed
H3: Anon IP scope	T1090.003	SigninLogs	5 (known)	No additional scope — matches incident
H3b: Lateral accts	T1078.004	SigninLogs	0 new	No lateral movement to other accounts
H4: Persistence	T1053/T1136	SecurityEvent+Sysmon	0	Cannot confirm or deny — no telemetry

4. Gap Analysis

Gap 1 — AMA Endpoint Telemetry Pipeline (Critical)	4 of 5 simulated ATT&CK; techniques were invisible to Sentinel and retrospective hunting. ATLAS rules 4.1 (brute force) and 4.2 (encoded PowerShell) were both non-functional. Diagnosis: check Heartbeat table, verify DCR resource association, re-register AMA agent. Preferred remediation: migrate to Microsoft Defender for Endpoint for always-on telemetry.
Gap 2 — No Phishing Click Telemetry in SIEM (Medium)	M365 Attack Simulation Training captured the click/credential result in the M365 Defender portal but did not forward this to Log Analytics. Real phishing campaigns generate EmailEvents in Advanced Hunting — simulation results are intentionally excluded. Enable M365 Defender for Office 365 connector in Sentinel content hub for real phishing detection.
Gap 3 — No SOAR Automation on Entra ID Protection Alerts (Medium)	When Entra ID Protection fired at 23:14, no playbook triggered to notify on-call, disable the account, or block sign-in. A Sentinel Logic App playbook with 'When an incident is created' trigger could reduce the 35-hour detection-to-triage gap to under 3 minutes.

5. ATT&CK; Coverage Assessment

Technique	Description	Detection	Hunt Coverage
T1110.001	Brute Force: Password Guessing	■ Not detected (AMA gap)	■ No data
T1059.001	PowerShell execution	■ Not detected (AMA gap)	■ No data
T1027	Obfuscated command	■ Not detected (AMA gap)	■ No data
T1566.002	Spearphishing Link	■■ M365 Defender only	■ Not in SIEM
T1078.004	Valid Accounts: Cloud	■ Entra ID Protection	■ No extra scope
T1090.003	Proxy: Multi-hop	■ Anonymous IP alert	■ No extra scope

Detection coverage: 2/6 techniques (33%) — entirely cloud identity layer.

6. Hunt Closure & Next Steps

- Hunt status: CLOSED — telemetry gap confirmed, cloud identity scope exhausted
- Open action: Remediate AMA agent on DESKTOP-TRF9U79 (P1 — Critical, see PIR)
- Next hunt trigger: After AMA remediation — repeat H1–H4 to validate detection coverage
- Recommended: Add weekly scheduled SigninLogs hunt for anomalous sign-in geography/velocity
- Hunt log retained: law-soc-atlas → Log Analytics → Saved Queries